

Trevor S. C. Johnson

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Objective

Adaptable and collaborative student seeking an internship position to leverage and improve my ability to contribute effectively to projects. Eager to apply my technical skills and problem-solving abilities to add value and support projects concerning VLSI and Semiconductor Engineering.

Seeking internship summer 2026 or full-time summer 2027.

Education

Georgia Institute of Technology | Atlanta, GA

Present

Bachelor of Science in Electrical Engineering, GPA 2.5

Expected Graduation, May 2027

Minor in Quantum Sciences and Technology

Skills

Programming: C++, C, Verilog, MATLAB, PLC Ladder Logic

Hardware: FPGAs, oscilloscope, logic analyzer, IC Tester, SMT PCB Machines, Raspberry Pi

Software: Cadence, Synopsys, Altera Quartus II, NI LabVIEW, SolidWorks, MATLAB

Communication: Design proposals, verification reports, technical reports, instruction manuals, presentations

Experience

Amazon Lab126 | Sunnyvale, CA

May – August 2022

Semiconductor Engineering Intern

- Worked with multiple teams taking a SoC for a potential consumer device from Architecture definition to Physical design
- Led a design verification phase, achieving full Register Transfer Level code coverage, improving overall design reliability

ExxonMobil | Houston, TX

Jan – Mar 2020

Digital Transformation Intern

- Developed APIs by analyzing distributor data to interface their ERPs with our server-side platforms

ExxonMobil | Baton Rouge, LA

May – Aug 2019

Instrumentation Engineer

- Designed industrial automation systems and controls using programmable logic controllers (PLCs)
- Specified control valves and transmitters to ensure safe, efficient, and reliable process control

Projects

Quantum Engineering

Fall Semester 2024 – Current

Vertically Integrated Product (VIP)

- Spearheads the interdisciplinary application of quantum technologies in engineering projects
- Comparing superconducting qubits and identifying novel modifications to enhance isolation and scalability in QC

Electronic Design Automation using Machine Learning

Fall Semester 2022 – Current

Personal project

- Received \$3000 from Google and \$2500 from AWS in the form of cloud research credits
- Integrated ML at a deep level in open-source EDA software (Yosys, OpenROAD, Magic) for design optimization
- Used a Transformer model and a generative adversarial network (GAN)

Chip Scale Power & Energy

Spring Semester 2023

Vertically Integrated Product (VIP)

- Developed and characterized chip-scale supercapacitors using functionalized carbon nanotube electrodes for advanced energy storage, integrating into sensors and IoT devices
- Designed a PCB for CubeSat applications, connecting supercapacitors to solar panels and devices for in-space applications

Leadership

The Hive (ECE Maker Space) | Atlanta, GA

Mar – Aug 2019

Vice President

- Secured funding and managed communications with corporate partners, to update and maintain equipment

ExxonMobil | Baton Rouge, LA

May – Aug 201

Wireless Instrumentation Project Lead

- Independently lead planning and design of wireless infrastructure implementation in the Alkylation Unit of the refinery
- Collaborated with contractors to determine optimal gateway and repeater locations